IT 360 FINAL REPORT

Introduction:

We developed a Python-based digital forensics tool that can be used to automate the extraction and organization of metadata from the digital evidence file. Metadata, such as file creation and modification times, are important for reconstructing timelines and identifying suspicious activity. This tool scans directories, extracts relevant metadata, and exports the results into a structured CSV format. This can help investigators to analyze evidence and detect anomalies across Linux and macOS systems. Our goal was to have a finished product that professionals may use for quick response.

Technical Implementation:

Program language: We used Python due to its simplicity, flexibility, and strong support for file handling and data processing. When it came to GitHub, we opted to share a word document with all directions. Our group decided to write out all the directions then have Sarah shift them into the GitHub. Even with all of us working together, most of the GitHub reports will show her making most of the commits- not reflecting the work we have all split up.

Libraries Used:

- os – Used for directory traversal and file path handling
- csv – Used to generate structured output files
- hashlib – Used to compute SHA256 hashes for file integrity verification
- datetime – Used for timestamp conversion and comparison
- Pillow (PIL) – Used to extract EXIF metadata from image files
- PyPDF2 – Included for potential PDF metadata extraction

Artificial Intelligence: For the AI aspect of this project, we planned on using Sushi to aid the user in understanding the information given. When the Sushi server shut down, we opted to use ai-assisted anomaly detection. This is using artificial intelligence and machine learning to analyze large, complex datasets and identify unusual patterns that deviate from established "normal" behavior. This was a perfect fit for our project, because are using a tool that may process large amounts of data.

Techniques used

1. Recursive File Scanning:

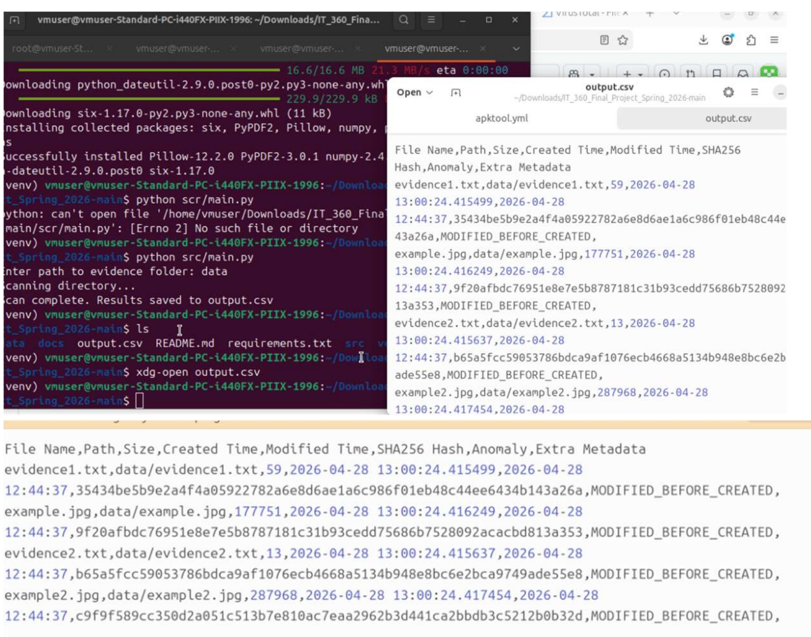
The tool scans all files in a directory and its subdirectories automatically

2. Metadata Extraction:
   Extracts file details such as name, size, and creation/modification times
3. File Hashing:
   Uses SHA256 to generate unique file fingerprints for integrity checks
4. EXIF Extraction:
   Reads embedded metadata from image files using Pillow
5. Anomaly Detection:
   Flags with unusual patterns like invalid timestamps or modified file history
6. CSV Output:
   Stores all result in a structured CSV file for easy analysis

Results:

Our results went just as we would hope, with having our tool fun on 3 different operating machines. Not only can our tool run through Linux, but it can also be used with macOS and Windows cmd. With the help of AI- we have a tool that can cut down on the search time of forensic professionals. Here are some of the outcomes, along with pictures, from our working tool:

Ubuntu:



Mac:

```
~/Downloads/IT_360_Final_Project_Spring_2026-main — -zsh

sofiahartmann@Sofias-MacBook-Air-5 IT_360_Final_Project_Spring_2026-main % python3 -m pip install -r requirements.txt
Collecting pandas (from -r requirements.txt (line 1))
  Downloading pandas-3.0.2-cp312-cp312-macosx_11_0_arm64.whl.metadata (79 kB)
Requirement already satisfied: Pillow in /Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages (from -r requirements.txt (line 2)) (12.2.0)
Collecting PyPDF2 (from -r requirements.txt (line 3))
  Downloading pypdf2-3.0.1-py3-none-any.whl.metadata (6.8 kB)
Requirement already satisfied: numpy>=1.26.0 in /Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages (from pandas->-r requirements.txt (line 1)) (2.2.3)
Collecting python-dateutil>=2.8.2 (from pandas->-r requirements.txt (line 1))
  Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting six>=1.5 (from python-dateutil>=2.8.2->pandas->-r requirements.txt (line 1))
  Using cached six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Downloading pandas-3.0.2-cp312-cp312-macosx_11_0_arm64.whl (9.9 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 9.9/9.9 MB 16.7 MB/s  0:00:00
Downloading pypdf2-3.0.1-py3-none-any.whl (232 kB)
Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
Installing collected packages: six, PyPDF2, python-dateutil, pandas
Successfully installed PyPDF2-3.0.1 pandas-3.0.2 python-dateutil-2.9.0.post0 six-1.17.0

[notice] A new release of pip is available: 26.0.1 -> 26.1
[notice] To update, run: pip install --upgrade pip
sofiahartmann@Sofias-MacBook-Air-5 IT_360_Final_Project_Spring_2026-main % python3 src/main.py
Enter path to evidence folder: data
Scanning directory...
Scan complete. Results saved to output.csv
sofiahartmann@Sofias-MacBook-Air-5 IT_360_Final_Project_Spring_2026-main % ls
data                docs                output.csv          README.md           requirements.txt        src
sofiahartmann@Sofias-MacBook-Air-5 IT_360_Final_Project_Spring_2026-main % cat output.csv
File Name,Path,Size,Created Time,Modified Time,SHA256 Hash,Anomaly Report,Extra Metadata
evidence1.txt,data/evidence1.txt,59,2026-04-28 13:33:36.771720,2026-04-28 11:30:54,35434be5b9e2a4f4a05922782a6e8d6ae1a6c986f01eb48c44ee6434b143a26a,Modified before created,
evidence2.txt,data/evidence2.txt,13,2026-04-28 13:33:36.771736,2026-04-28 11:30:54,b65a5fcc59053786bdca9af1076ecb4668a5134b948e8bc6e2bca9749ade55e8,Modified before created,
example2.jpg,data/example2.jpg,287968,2026-04-28 13:33:36.771770,2026-04-28 11:30:54,c9f9f589cc350d2a051c513b7e810ac7eaa2962b3d441ca2bbdb3c5212b0b32d,Modified before created,
example.jpg,data/example.jpg,177751,2026-04-28 13:33:36.771753,2026-04-28 11:30:54,9f20afbdc76951e8e7e5b8787181c31b93cedd75686b7528092acacbd813a353,Modified before created,
sofiahartmann@Sofias-MacBook-Air-5 IT_360_Final_Project_Spring_2026-main %
```

Windows:

File Name	Path	Size	Created Ti	Modified T	SHA256 Ha	Anomaly R	Extra Metadata
evidence1	C:\Users\c	59	########	07:07.1	35434be5b	NORMAL	
evidence2	C:\Users\c	13	########	07:07.1	b65a5fcc5	NORMAL	
example.jp	C:\Users\c	177751	########	07:07.2	9f20afbdc7	NORMAL	
example2.	C:\Users\c	287968	########	07:07.2	c9f9f589c	NORMAL	

```
File Name,Path,Size,Created Time,Modified Time,SHA256 Hash,Anomaly Report,Extra Metadata
evidence1.txt,C:\Users\criol\Downloads\IT_360_Final_Project_Spring_2026-main\IT_360_Final_Project_Spring_2026-main
\data\evidence1.txt,59,2026-04-28 16:02:38,2026-04-28
18:07:07.126128,35434be5b9e2a4f4a05922782a6e8d6ae1a6c986f01eb48c44ee6434b143a26a,NORMAL,
evidence2.txt,C:\Users\criol\Downloads\IT_360_Final_Project_Spring_2026-main\IT_360_Final_Project_Spring_2026-main
\data\evidence2.txt,13,2026-04-28 16:02:38,2026-04-28
18:07:07.147246,b65a5fcc59053786bdca9af1076ecb4668a5134b948e8bc6e2bca9749ade55e8,NORMAL,
example.jpg,C:\Users\criol\Downloads\IT_360_Final_Project_Spring_2026-main\IT_360_Final_Project_Spring_2026-main
\data\example.jpg,177751,2026-04-28 16:02:38,2026-04-28
18:07:07.166799,9f20afbdc76951e8e7e5b8787181c31b93cedd75686b7528092acacbd813a353,NORMAL,
example2.jpg,C:\Users\criol\Downloads\IT_360_Final_Project_Spring_2026-main\IT_360_Final_Project_Spring_2026-main
\data\example2.jpg,287968,2026-04-28 16:02:38,2026-04-28
18:07:07.181445,c9f9f589cc350d2a051c513b7e810ac7eaa2962b3d441ca2bbdb3c5212b0b32d,NORMAL,
```

```
PS C:\Users\criol\Downloads\IT_360_Final_Project_Spring_2026-main\IT_360_Final_Project_Spring_2026-main\src> Get-Content
 output.csv
File Name,Path,Size,Created Time,Modified Time,SHA256 Hash,Anomaly Report,Extra Metadata
evidence1.txt,C:\Users\criol\Downloads\IT_360_Final_Project_Spring_2026-main\IT_360_Final_Project_Spring_2026-main\data\
evidence1.txt,59,2026-04-28 16:02:38,2026-04-28 18:07:07.126128,35434be5b9e2a4f4a05922782a6e8d6ae1a6c986f01eb48c44ee6434
b143a26a,NORMAL,
evidence2.txt,C:\Users\criol\Downloads\IT_360_Final_Project_Spring_2026-main\IT_360_Final_Project_Spring_2026-main\data\
evidence2.txt,13,2026-04-28 16:02:38,2026-04-28 18:07:07.147246,b65a5fcc59053786bdca9af1076ecb4668a5134b948e8bc6e2bca974
9ade55e8,NORMAL,
example.jpg,C:\Users\criol\Downloads\IT_360_Final_Project_Spring_2026-main\IT_360_Final_Project_Spring_2026-main\data\ex
ample.jpg,177751,2026-04-28 16:02:38,2026-04-28 18:07:07.166799,9f20afbdc76951e8e7e5b8787181c31b93cedd75686b7528092acacb
d813a353,NORMAL,
example2.jpg,C:\Users\criol\Downloads\IT_360_Final_Project_Spring_2026-main\IT_360_Final_Project_Spring_2026-main\data\e
xample2.jpg,287968,2026-04-28 16:02:38,2026-04-28 18:07:07.181445,c9f9f589cc350d2a051c513b7e810ac7eaa2962b3d441ca2bbdb3c
5212b0b32d,NORMAL,
PS C:\Users\criol\Downloads\IT_360_Final_Project_Spring_2026-main\IT_360_Final_Project_Spring_2026-main\src>
```

Lessons Learned & Conclusion:

This project taught us a lot more than we expected going in. The biggest struggle we ran into was getting the tool to run properly across different operating systems. MacOS and Linux gave us their fair share of headaches with dependencies and environment setup, but

Windows was by far the hardest to get working. The file paths are formatted differently, Python must be configured in a specific way, and the terminal behaves like it does on a Mac or Linux machine. Since Sushi was no longer an option to use within our project, the inclusion of AI was difficult for our group. Getting the AI anomaly detection to work was challenging because we had to figure out what counts as suspicious behavior versus something totally normal, which is harder to define than it sounds. On the forensics side, this project really opened our eyes to how much information is hidden inside everyday files. Metadata tells a story, and learning how investigators use things like timestamps, file hashes, and EXIF data to piece together what happened to a file was fascinating. As a group, we were able to learn that newer versions of Ubuntu require using a virtual environment, known as venv, to install Python libraries, which was necessary for managing dependencies properly. Overall, we walked away from this project with a much better understanding of digital forensics and a lot more respect for the people who do this kind of work professionally. We hope that our tool can help out someone looking to use it as a forensics aid, or someone who is just looking to learn more about the digital forensics behind it.